recordings (also discussed in Chapter 17). These types of linear techniques are the backbone of DSP. Several *nonlinear* techniques are also useful for audio processing. Two will be briefly described here.

The first nonlinear technique is used for reducing wideband noise in speech signals. This type of noise includes: magnetic tape hiss, electronic noise in analog circuits, wind blowing by microphones, cheering crowds, etc. Linear filtering is of little use, because the frequencies in the noise completely overlap the frequencies in the voice signal, both covering the range from 200 hertz to 3.2 kHz. How can two signals be separated when they overlap in both the time domain *and* the frequency domain?

Here's how it is done. In a short segment of speech, the amplitude of the frequency components are greatly *unequal*. As an example, Fig. 22-10a illustrates the frequency spectrum of a 16 millisecond segment of speech (i.e., 128 samples at an 8 kHz sampling rate). Most of the signal is contained in a few large amplitude frequencies. In contrast, (b) illustrates the spectrum when only random noise is present; it is very irregular, but more uniformly distributed at a low amplitude.

Now the key concept: if both signal and noise are present, the two can be partially separated by looking at the *amplitude* of each frequency. If the amplitude is large, it is probably mostly signal, and should therefore be retained. If the amplitude is small, it can be attributed to mostly noise, and should therefore be discarded, i.e., set to zero. Mid-size frequency components are adjusted in some smooth manner between the two extremes.

Another way to view this technique is as a *time varying Wiener filter*. As you recall, the frequency response of the Wiener filter passes frequencies that are mostly signal, and rejects frequencies that are mostly noise. This

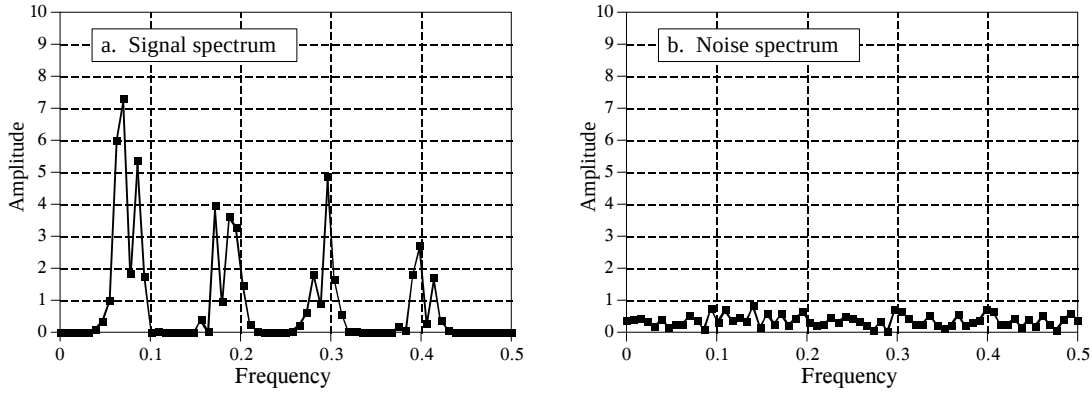

FIGURE 22-10
Spectra of speech and noise. While the frequency spectra of speech and noise generally overlap, there is some separation if the signal segment is made short enough. Figure (a) illustrates the spectrum of a 16 millisecond speech segment, showing that many frequencies carry little speech information, *in this particular segment*. Figure (b) illustrates the spectrum of a random noise source; all the components have a small amplitude. (These graphs are not of real signals, but illustrations to show the noise reduction technique).